VNR VJIET

Name of the Laboratory:

Name of the Experiment:

Experiment No: **Date:**

INTRODUCTION TO OPERATING SYSTEM FUNDAMENTALS

1)What are the types of operating systems?

Ans: Types of operating systems are:

1. Batch operating System

A Batch Operating System is designed to handle large groups of similar jobs efficiently. It does not interact with the computer directly but instead processes jobs that are grouped by an operator. These jobs are queued and executed one after the other, without user interaction during the process.

2. Multi-Programming Operating System

In a Multi-Programming Operating System, multiple programs run in memory at the same time. The CPU switches between programs, utilizing its resources more effectively and improving overall system performance.

3. Multi-tasking/Time-sharing Operating systems

Multitasking OS is a type of multiprogramming system where each process runs in a round-robin manner. Every task gets a fixed time slice called a quantum. After the quantum ends, the OS switches to the next task, allowing multiple tasks whether from one user or many to run smoothly on a single system.

4. Multi-Processing Operating System

Multi-Processing OS is a type of Operating System in which more than one CPU is used for the execution of processes. It betters the throughput of the System.

5. Distributed Operating System

Distributed operating systems connects multiple independent computers through a shared communication network. Each system has its own CPU and memory but works together as a single unit. The main benefit is remote access, allowing users to use files and software stored on other connected systems.

6. Network Operating System

A Network Operating System (NOS) runs on a server and manages data, users, security, applications, and other network functions. It allows shared access to files, printers, and resources within a small private network. Users can see the configuration and connections of other users, which is why these systems are considered tightly coupled systems.

7. Real-Time Operating System

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time. Real-time systems** are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

8. Mobile Operating Systems

Mobile operating systems are designed specifically for mobile devices such as smartphones and tablets. Examples of such operating systems are Android and iOS. These operating systems manage the hardware and software resources of the device, providing a platform for running applications and ensuring a seamless user experience.

2. Give examples for each OS specified above.

Ans: EXAMPLES:

1. Batch Operating System

- Banking systems
- Railway servers
- Billing machines

2. Multi-Programming Operating System

- Banking systems
- Railway servers
- Billing machines

3. Multi-tasking/Time-sharing Operating systems

- IBM VM/CMS
- TSO (Time Sharing Option)
- Windows Terminal Services

4. Multi-Processing Operating System

- UNIX
- Linux (Ubuntu, Red Hat, Debian)
- macOS

5. Distributed Operating System

- LOCUS
- MICROS
- Amoeba

6. Network Operating System

- Microsoft Windows Server 2003
- UNIX, Linux
- Mac OS X

7. Real-Time Operating System

- Scientific experiments
- Medical imaging systems
- Robots

8. Mobile Operating Systems

- Android
- iOS
- Blackberry

3. Draw one device for each type of operating system.

Ans: One device for each type of operating system

- Batch OS: Mainframe computer
- Multiprogramming OS: Desktop PC
- Multitasking OS: Laptop
- Multiprocessing OS: Server
- Real-Time OS: Car
- Distributed OS: Connected computers
- Network OS: Server with clients
- Embedded OS: Washing machine
- Mobile OS: Smartphone

4. List out the general types of operating systems.

Ans: The general types of operating systems are:

1. Batch Operating System
2. Multiprogramming Operating System
3. Multitasking (Time-Sharing) Operating System
4. Multiprocessing Operating System
5. Real-Time Operating System (RTOS)
6. Distributed Operating System
7. Network Operating System (NOS)
8. Embedded Operating System
9. Mobile Operating System

5.Evaluation of Windows operating system.

Ans: Evolution of the Windows Operating System with the details of all the versions from 1985 till the present :

- An operating system is software that is responsible for the communication between the user and the hardware of the computer.
- Windows is the most popular operating system in the world.
- Windows is developed by Microsoft Corporation in 1985 and is also known as Microsoft Windows.
- Windows 1 was the first version of the Windows operating system released by Microsoft designed using the Graphical User Interface (GUI) and did not support overlapping windows.
- Windows 2 was released by Microsoft on December 9, 1987. It allowed the applications to overlap with each other.

- Windows 3 was released by Microsoft on May 22, 1990. Multitasking was supported in this version which was a major advancement.
- Windows 95 was released by Microsoft on August 15, 1995. It introduced the Taskbar and the Start Menu and it advanced from a 16-bit GUI to a 32-bit GUI.
- Windows 98 was released by Microsoft on June 25, 1998. It was released as an upgraded version of Windows 95 and it supported multiple monitors on one computer and internet sharing.
- The Windows 2000 was released by Microsoft on February 17, 2000. It supports preemptive multitasking and has a Windows file protection system.
- Windows XP was released by Microsoft on October 25, 2001. It has advanced portable PC support.

- Windows Vista was released by Microsoft on January 30, 2007. The security control feature known as the User Account Control (UAC) was responsible for blocking unauthorized OS changes.
- Windows 7 was released by Microsoft on October 22, 2009. It was one of the best versions of the Windows operating system. The improved cryptography enhanced BitLocker's capabilities in terms of security.
- Windows 8 was released by Microsoft on October 26, 2012. The security changes done for this version were mostly hardware based.
- Windows 10 was released by Microsoft on July 29, 2015. Cortana - The virtual assistant was a new feature in the Windows 10 OS.

6. Distributions of Linux operating system.

Ans: A Linux distribution is an operating system built on the Linux kernel, bundled with essential software, tools, and package managers, customized to serve different users like developers, enterprises, cybersecurity professionals, and general users. They are:

- **Debian**
- **Ubuntu**
- **Linux Mint**
- **MX Linux**
- **Zorin OS**
- **Pop!_OS**
- **Fedora**
- **Red Hat Enterprise Linux (RHEL)**
- **CentOS Stream**
- **Rocky Linux**
- **AlmaLinux**
- **Arch Linux**
- **Manjaro**

- **Garuda Linux**
- **EndeavourOS**
- **openSUSE Leap**
- **openSUSE Tumbleweed**
- **Gentoo**
- **Slackware**
- **Kali Linux**
- **Tails**
- **Elementary OS**

- **Alpine Linux**

7.In high-end servers, which operating system is used and why?

Ans: High-end servers typically use Linux/UNIX-based operating systems (such as Ubuntu Server, Red Hat Enterprise Linux, SUSE Linux Enterprise, Debian, or UNIX variants). Because:

- High stability and reliability (can run continuously with minimal downtime)
- Better security against threats
- Efficient resource management
- Excellent support for multi-user and multitasking environments
- High performance and scalability
- Strong networking capabilities
- Cost-effective (many Linux distributions are open source)
- Widely used for cloud computing, web hosting, databases, and enterprise applications